

PROFESSIONAL SUMMARY

Data Analyst with 4+ years of experience in delivering impactful data-driven solutions across diverse industries, including healthcare & finance. Expertise in data analysis, visualization & predictive modeling utilizing a comprehensive suite of tools including SQL, Python (Pandas, NumPy, Scikit-learn), Tableau, Qlik Sense, Power BI, SSIS, Apache Airflow, Azure Cloud, Snowflake, AWS S3, Apache Hive, Apache Spark, Jira & Confluence. Proven ability to effectively collaborate with cross-functional teams, manage projects using Agile methodologies & communicate complex analytical findings to technical & non-technical audiences.

TECHNICAL SKILLS

Data Visualization: Tableau, Qlik Sense, Power BI, Matplotlib, Seaborn

Data Analysis: Python (Pandas, NumPy, Scikit-learn), SQL, Excel (Advanced Functions, VBA Macros)

Machine Learning: Scikit-learn, Regression Models, Clustering Techniques (K-means), Predictive Modeling

ETL and Data Engineering: SSIS, Apache Airflow, SQL Server, Alteryx, AWS S3, Apache Hive, Apache Spark, Snowflake

Cloud Platforms: AWS (EC2, S3), Azure Cloud Infrastructure

Data Governance: HIPAA Compliance, Data Encryption Protocols, Data Quality Audits

Big Data: Apache Hive, Apache Spark

Database Management: SQL, PostgreSQL, Snowflake, Data Lake Architectures

Programming Languages: Python, SQL

Statistical Analysis: A/B Testing, Root Cause Analysis, Time-Series Analysis, Hypothesis Testing, Regression Analysis

Data Integration: APIs, Electronic Health Records (EHR), Claims Databases, ICD, HL7

Project Management: Jira, Confluence, Agile, Scrum

Development & Collaboration: Git, GitHub

Reporting Tools: Excel (Advanced Functions, VBA Macros), Tableau, Power BI, Qlik Sense

Other Tools & Frameworks: MS Visio, Alteryx, SSIS, Confluence

Operating Systems: Windows, Linux, Mac iOS

EDUCATION

Masters in Data Science - New Jersey Institute of Technology, Newark, NJ, USA

Bachelors in Electronics and communications engineering - K.S.R.M College of Engineering Andhra Pradesh, India

PROFESSIONAL EXPERIENCE

UnitedHealth Group – VA | Data Analyst | Sep 2023 - Present

- Designed and developed dynamic dashboards in Tableau and Qlik Sense to monitor key performance indicators (KPIs) and ensure compliance with Medicare and Medicaid regulations, improving decision-making efficiency by 25%.
- Engineered robust ETL pipelines using SSIS & Apache Airflow to consolidate data from Electronic Health Record (EHR) systems, claims databases & external APIs, ensuring data accuracy and completeness.
- Performed advanced data cleaning, transformation & preprocessing on datasets exceeding millions of rows using Python libraries (Pandas, NumPy) & optimized SQL queries, reducing processing time by 40%.
- Implemented stringent data governance practices by conducting regular audits, applying encryption protocols and ensuring compliance with HIPAA standards, minimizing data breach risks.
- Collaborated with cross-disciplinary teams to align data and reporting structures with ICD and HL7 standards, facilitating interoperability and seamless data exchange between systems.
- Developed predictive machine learning models using Scikit-Learn to identify patients at high risk of readmission, achieving a precision rate of 85% and reducing readmission rates by 20%.
- Created time-series analysis models to evaluate trends in claim submissions and reimbursement delays, enabling proactive strategies to reduce claim processing times by 15%.
- Automated complex ad-hoc reporting tasks, leveraging Python scripting, which decreased manual reporting efforts by 50% and improved turnaround time for critical reports.
- Analyzed and visualized patient flow data to identify bottlenecks in emergency room operations, implementing data-driven recommendations that improved throughput by 15%.
- Built classification and clustering models to segment patient populations and identify high-risk groups, enabling targeted interventions and personalized care plans.
- Maintained and optimized data pipelines on Azure Cloud Infrastructure with integrated Snowflake for real-time analytics, achieving 99% system uptime and seamless data workflows.
- Utilized Agile methodologies, managing project deliverables through Jira and documenting requirements and sprint outcomes on Confluence, ensuring timely delivery of data analytics solutions.

HCLTech – India | Data Analyst | Jun 2019 – Jul 2022

- Deployed a sophisticated predictive credit risk scoring model utilizing Python and machine learning algorithms, resulting in a 20% enhancement in predictive accuracy over legacy models, thereby improving loan approval processes.
- Implemented dynamic, user-friendly dashboards in Power BI that streamlined data visualization & reporting processes, achieving a 30% reduction in report generation time & enabling stakeholders to make data-driven decisions swiftly.
- Crafted and optimized complex SQL queries for efficient extraction, transformation and loading (ETL) of large financial datasets, ensuring high performance and minimal latency in data retrieval operations.
- Conducted rigorous A/B testing to assess the impact of innovative financial strategies, delivering actionable insights that informed executive decision-making & enhanced overall business performance.
- Engineered advanced machine learning models specifically for fraud detection, resulting in a 15% improvement in detection rates and significantly reducing financial losses due to fraudulent activities.
- Maintained robust ETL pipelines using SQL Server Integration Services (SSIS) & Python, ensuring seamless data integration from multiple sources & enhancing data quality for analytical purposes.
- Generated detailed financial reports and performed comprehensive root cause analysis on data anomalies, providing critical insights that supported strategic planning and operational efficiency.
- Leveraged Apache Hive and Apache Spark for large-scale data processing and analytics, enabling the handling of massive datasets with agility and precision, thereby enhancing the organization's analytical capabilities.
- Construct sophisticated financial risk models to evaluate market and credit risks, significantly improving the organization's risk management frameworks and compliance with regulatory standards.
- Implemented scalable data lake architectures on AWS S3, optimizing data storage solutions and facilitating efficient data retrieval processes for analytics and reporting.
- Collaborated effectively with cross-functional teams using JIRA for Agile project management, ensuring alignment on project goals and timely delivery of analytical solutions.
- Actively participated in Agile and Scrum methodologies, contributing to team sprints, backlog grooming and project planning sessions, fostering a culture of continuous improvement and adaptability.